hook of (i,j) to be the set of all squares (i',j') in λ with $i' \geq i$, $j' = j$ or $i' = i$, $j' \geq j$. Let $h(i,j)$ be the length of the hook of i,j, i.e., the number of squares in it.

Theorem 5.17.1 (The hook length formula)**.** *One has*

$$\dim V_\lambda = \frac{n!}{\prod_{(i,j):i\leq\lambda_j} h(i,j)}.$$

Proof. The formula follows from formula (5.17.1). Namely, note that

$$\frac{l_1!}{\prod_{1<j\leq N}(l_1-l_j)} = \prod_{1\leq k\leq l_1, k\neq l_1-l_j} k.$$

It is easy to see that the factors in this product are exactly the hook lengths $h(i,1)$. Now delete the first row of the diagram and proceed by induction. □

5.18. Schur-Weyl duality for $\mathfrak{gl}(V)$

We start with a simple result which is called the **Double Centralizer Theorem**.

Theorem 5.18.1. *Let A, B be two subalgebras of the algebra* $\operatorname{End} E$ *of endomorphisms of a finite dimensional vector space E, such that A is semisimple and $B = \operatorname{End}_A E$. Then:*

(i) $A = \operatorname{End}_B E$ (i.e., the centralizer of the centralizer of A is A).

(ii) B is semisimple.

(iii) As a representation of $A \otimes B$, E decomposes as

$$E = \bigoplus_{i\in I} V_i \otimes W_i,$$

where V_i are all the irreducible representations of A and W_i are all the irreducible representations of B. In particular, we have a natural bijection between irreducible representations of A and B.

Proof. Since A is semisimple, we have a natural decomposition $E = \bigoplus_{i\in I} V_i \otimes W_i$, where $W_i := \operatorname{Hom}_A(V_i, E)$ and $A = \bigoplus_i \operatorname{End} V_i$. Note that $W_i \neq 0$, since the action of A on E is faithful. Therefore, by